COPD and Myocardial Infarction/Coronary Artery Disease: Correlation and Causation

1. Introduction

Chronic obstructive pulmonary disease (COPD) and coronary artery disease (CAD) represent two of the world's leading causes of morbidity and mortality, and their frequent co-occurrence has generated intense research interest regarding whether the relationship extends beyond shared risk factors to genuine causal interaction (Boschetto et al., 2012; Roversi et al., 2014). Cardiovascular disease accounts for up to one-third of all deaths in COPD patients, making cardiac disease — particularly ischemic heart disease and acute myocardial infarction (AMI) — the leading non-respiratory cause of mortality in this population (Goedemans et al., 2020; Brassington et al., 2022; Polman et al., 2024). The prevalence of COPD among AMI populations ranges from 7% to 30%, while the prevalence of ischemic heart disease in COPD patients ranges from 4% to 60% depending on the population studied and diagnostic methods used (Goedemans et al., 2020; Roversi et al., 2014; Polman et al., 2024).

A central question is whether COPD independently causes or accelerates coronary artery disease and myocardial infarction, or whether the observed associations are confounded by shared risk factors — principally smoking and aging (Goedemans et al., 2020; Rothnie et al., 2015). Multiple large observational studies and meta-analyses have demonstrated that COPD roughly doubles the risk of myocardial infarction even after adjustment for traditional cardiovascular risk factors (Rothnie et al., 2015; Ragnoli et al., 2025; Amiri et al., 2025), and a Mendelian randomization study has attempted to disentangle causation from confounding using genetic instruments (Yu et al., 2024). The mechanistic literature points to systemic inflammation, oxidative stress, hypoxia, endothelial dysfunction, arterial stiffness, and hypercoagulability as COPD-specific pathways that may drive atherosclerosis and plaque rupture independently of smoking (Goedemans et al., 2020; Daher & Dreher, 2020; Ragnoli et al., 2025; Amiri et al., 2025). Acute exacerbations of COPD have emerged as periods of particularly heightened cardiovascular risk, with myocardial infarction incidence spiking in the days following severe exacerbations (Donaldson et al., 2010; Rothnie et al., 2018; Wallström et al., 2024).

Does COPD independently increase the risk of myocardial infarction and coronary artery disease?

Requires at least 5 papers that directly answer your question. Try adjusting your query to find more papers.

FIGURE 1 Research consensus on COPD as an independent cardiovascular risk factor

2. Methods

The search ran over 170 million research papers indexed in Consensus, drawing from Semantic Scholar, PubMed, and other biomedical databases. An initial broad search for the correlation and causation between COPD and myocardial infarction/coronary artery disease yielded over 1 million results. Six targeted research groups explored foundational concepts, causality, terminology variants, contrasting findings, related mechanisms, and adjacent topics including endothelial dysfunction, oxidative stress, and heart failure. Citation crawling from 49 seed papers identified 1,389 additional related works. Machine-learned relevance filtering screened 229 papers, of which 136 passed the relevance threshold. After deduplication, the top 50 papers were selected by relevance ranking for this synthesis.

Search Strategy

79.1M Retrieved	→	1.5K Eligible	→	50 Included
21 searches 1 citation graph use All papers retrieved from 21 searches and 1 citation graph use.		Papers meeting relevance and quality criteria, after deduplication		Top papers selected to be included in the final analysis

FIGURE 2 Search strategy and paper selection flow diagram

The strategy combined broad epidemiological searches with mechanistic and causal-inference queries to capture the full spectrum of evidence.

3. Results

3.1 Key Papers

The corpus is anchored by several foundational works. Donaldson et al. (2010), cited 536 times, established the temporal link between COPD exacerbations and acute MI using a self-controlled case series design in 25,857 patients (Donaldson et al., 2010). Rothnie et al. (2015), a systematic review and meta-analysis cited 103 times, quantified the pooled risk of MI in COPD and identified smoking as a critical confounder (Rothnie et al., 2015). The Mendelian randomization study by Yu et al. (2024) represents the most direct attempt to distinguish causation from confounding using genetic instruments (Yu et al., 2024).

Paper	Summary
(Seleno, 2011)	Exacerbations trigger acute MI within 5 days (Donaldson et al., 2010)
(Polman et al., 2024)	Meta-analysis: COPD increases MI risk (HR 1.72) (Rothnie et al., 2015)
(Miklós & Horváth, 2023)	MR study: COPD–CHD causality mediated by confounders (Yu et al., 2024)

FIGURE 3 Key papers anchoring the COPD–CAD evidence base

3.2 Epidemiological Association

Multiple meta-analyses and large cohort studies converge on a robust association between COPD and increased cardiovascular risk. A meta-analysis of 6.4 million patients found that comorbid COPD-CAD patients had triple the all-cause mortality of non-COPD CAD patients (RR 2.81, 95% CI 2.40–3.29), with significantly higher rates of myocardial infarction, cardiac death, stroke, and heart failure (Zheng et al., 2024). A systematic review of 24 observational studies showed COPD was associated with increased MI risk in cohort analyses (HR 1.72, 95% CI 1.22–2.42), though the association was not significant in case-control studies (OR 1.18, 95% CI 0.80–1.76) (Rothnie et al., 2015). In a primary prevention cohort of 5.8 million individuals without prior CVD, those with COPD had a 25% higher rate of major adverse cardiovascular events after adjustment for cardiovascular risk factors (HR 1.25, 95% CI 1.23–1.27), a magnitude comparable to diabetes (Maclagan et al., 2023). Among people with established CVD, COPD was associated with a 24% higher rate of MACE (HR 1.24, 95% CI 1.21–1.26) after full adjustment (Cho et al., 2025). A nationwide Danish registry study of 1.47 million residents found COPD patients had higher absolute risks of MI across all age groups, with the youngest group (50–60 years) showing nearly double the MI risk of non-COPD counterparts (4.8% vs. 2.7%) (Lauridsen et al., 2025). In a UK primary care analysis of 1.2 million individuals, the hazard ratio for acute MI in COPD patients aged 35–44 was 10.34 (95% CI 3.28–32.60), with the greatest excess risk concentrated in younger age groups (Feary et al., 2010).

3.3 Causality: Mendelian Randomization and Confounding

The question of whether COPD causally drives coronary artery disease — rather than merely co-occurring due to shared risk factors — has been most directly addressed by a bidirectional Mendelian randomization study. Yu et al. found a statistically significant genetic causal association between COPD and coronary heart disease in univariate analysis (OR 1.004, 95% CI 1.002–1.006, $p < 0.001$), but this association was abolished after adjusting for the inflammatory mediator IL-6, LDL cholesterol, or total cholesterol ($p > 0.05$) (Yu et al., 2024). This suggests that the COPD–CHD causal pathway is substantially mediated by systemic inflammation and lipid metabolism rather than representing a direct independent effect (Yu et al., 2024). By contrast, the same study found that COPD had causal relationships with heart failure, essential hypertension, and stroke that persisted after multivariable adjustment (Yu et al., 2024). Observational evidence consistently shows that COPD confers cardiovascular risk beyond shared risk factors: reduced lung function has been identified as an independent predictor of cardiovascular mortality in the general population (Rothnie et al., 2015), and COPD has been characterized as a risk factor for CAD development independent of other cardiovascular risk factors (Daher & Dreher, 2020; Boschetto et al., 2012). However, a major limitation across the observational literature is inadequate adjustment for smoking status, which is the dominant shared confounder (Rothnie et al., 2015). One angiographic study even found a negative association between COPD and angiographically confirmed CAD (adjusted OR 0.75, 95% CI 0.67–0.84), potentially reflecting survivorship bias or referral patterns (Hong et al., 2019).

3.4 Mechanisms Linking COPD to Atherosclerosis and MI

Mechanism	Evidence Base	Key Finding	Citations
Systemic inflammation	Strong (multiple studies)	Elevated IL-6, CRP, fibrinogen drive atherosclerosis (Goedemans et al., 2020; Terrosu, 2020; Roversi et al., 2014)	(Goedemans et al., 2020)

Mechanism	Evidence Base	Key Finding	Citations
Oxidative stress	Moderate (reviews)	Amplifies inflammation and endothelial damage (Miklós & Horváth, 2023; Brassington et al., 2022)	(Miklós & Horváth, 2023)
Hypoxia	Moderate	Activates RAS, reduces renal blood flow, increases arterial stiffness (Goedemans et al., 2020; Amiri et al., 2025)	(Goedemans et al., 2020)
Hypercoagulability	Moderate	Increased platelet reactivity and thrombin levels (Goedemans et al., 2020)	(Goedemans et al., 2020)
Lung hyperinflation	Emerging	Strongest pulmonary predictor of clinical and subclinical CAD (Chandra et al., 2021; Polman et al., 2024)	(Chandra et al., 2021)
Endothelial dysfunction	Emerging	Early event in atherogenesis linking COPD to CVD (Marcuccio et al., 2025; Casselbrant et al., 2022)	(Marcuccio et al., 2025)

FIGURE 4 Mechanistic pathways linking COPD to coronary artery disease

Lung hyperinflation emerged as the pulmonary phenotype most strongly associated with clinical and subclinical CAD, with FEV1 and emphysema associations attenuated after adjusting for hyperinflation (Chandra et al., 2021). Matrix metalloproteinases elevated in COPD increase the risk of atherosclerotic plaque formation, destabilization, and rupture, and have been linked to left ventricular remodeling after AMI (Goedemans et al., 2020). Microvascular dysfunction has been proposed as a pivotal intermediary, with systemic inflammation and oxidative stress damaging the vascular endothelium and setting the stage for atherosclerosis and ischemic injury (Amiri et al., 2025).

Results Timeline

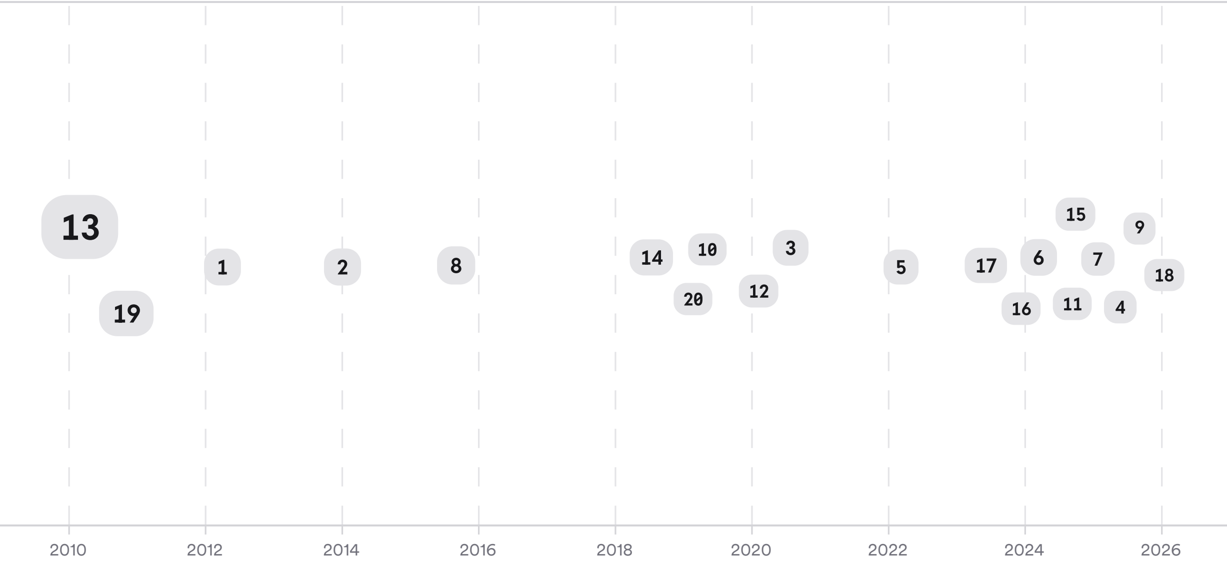

FIGURE 5 Timeline of influential COPD–cardiovascular disease research; larger markers indicate more citations

Research on the COPD–cardiovascular relationship has progressed through three phases. Early epidemiological studies (pre-2010) established the comorbidity burden and shared risk factors. From 2010–2018, self-controlled case series and large cohort studies quantified the temporal link between exacerbations and acute MI, with Donaldson et al. (2010) and Rothnie et al. (2018) as landmarks. From 2019 onward, mechanistic reviews, Mendelian randomization, and imaging studies have sought to disentangle causation from confounding and identify specific biological pathways.

Top Contributors

Type	Name	Papers
Author	J. Quint	[ce0549ca46585783acbaa78384ec143d][42cd99881ebc5e02840c822ec11ec858][53e44dfb6b4d5e31b5350204f1fe194a][5f4cefd9385154bc94dec28cf90647a7][8ba5373f1805539b9f4c802bd29725f3]
Author	L. Smeeth	[ce0549ca46585783acbaa78384ec143d][42cd99881ebc5e02840c822ec11ec858][5f4cefd9385154bc94dec28cf90647a7]
Author	L. Fabbri	[93a0699d1a01505bbacb0cc6496f5ddb][c5d271d6b0eb5b9d81d4152986e0e522]
Journal	*Thorax*	[d8d75da2f5115940bd7ecea275083590][8c83cf042bc951de83e23b38ebf21ca1]
Journal	*Chest*	[15e85bfe07dc572986876de0e58fdd57][3aa55dbf6d5b5ae19f42754c7914a829][0702480d76a45fafbc80a1d79e9404f9][a6682ed0f713507ba839345d3b9103be]

Type	Name	Papers
Journal	*BMJ Open*	[ce0549ca46585783acbaa78384ec143d]

FIGURE 6 Authors and journals that appeared most frequently in the included papers

4. Discussion

The observational evidence for an association between COPD and myocardial infarction/coronary artery disease is robust and replicated across multiple large cohorts, meta-analyses, and systematic reviews spanning diverse populations and healthcare systems (Rothnie et al., 2015; Zheng et al., 2024; Sá-Sousa et al., 2024; Maclagan et al., 2023). Effect sizes are clinically meaningful: COPD roughly doubles MI risk, and comorbid COPD-CAD patients face triple the mortality of CAD patients without COPD (Ragnoli et al., 2025; Zheng et al., 2024). The association persists after adjustment for traditional cardiovascular risk factors in most studies, supporting the interpretation that COPD-specific mechanisms contribute independently (Daher & Dreher, 2020; Maclagan et al., 2023; Cho et al., 2025).

However, the causal interpretation faces important limitations. The dominant confounder — smoking — is inadequately controlled in many studies, particularly those using administrative databases that do not record smoking status (Rothnie et al., 2015). The Mendelian randomization study by Yu et al. provides the most rigorous causal test and found that the COPD–CHD association was abolished after adjusting for IL-6 and lipid levels, suggesting mediation rather than direct causation (Yu et al., 2024). This finding aligns with the mechanistic literature positioning systemic inflammation as the central pathway: COPD-driven elevations in CRP, IL-6, TNF-α, and fibrinogen promote atherosclerosis, plaque instability, and thrombosis (Terrosu, 2020; Ferreira et al., 2023). The bidirectional nature of the relationship is also underexplored — whether MI triggers COPD exacerbations or vice versa remains difficult to disentangle given the cross-sectional nature of much evidence (De Molina et al., 2018).

Acute exacerbations represent the most compelling evidence for a dynamic causal link. The self-controlled case series design, which controls for time-invariant confounders within individuals, consistently shows that MI risk spikes 2.3- to 2.6-fold in the days after severe exacerbations and diminishes over time (Donaldson et al., 2010; Rothnie et al., 2018; Wallström et al., 2024). This temporal specificity — with risk peaking within 1–5 days and returning to baseline — strongly suggests that the inflammatory surge during exacerbations acutely destabilizes coronary plaques rather than reflecting chronic shared risk factors (Donaldson et al., 2010; Terrosu, 2020).

Claim	Evidence Strength	Reasoning	Papers
COPD is associated with increased MI risk after adjustment for traditional risk factors	Strong	Multiple meta-analyses and large cohorts; consistent direction and magnitude; residual confounding by smoking remains	(Rothnie et al., 2015; Maclagan et al., 2023; Cho et al., 2025; Zheng et al., 2024)
Acute COPD exacerbations transiently increase MI risk within days	Strong	Self-controlled case series design eliminates time-invariant confounding; replicated across two large UK cohorts with consistent effect sizes	(Donaldson et al., 2010; Rothnie et al., 2018; Wallström et al., 2024)

Claim	Evidence Strength	Reasoning	Papers
Systemic inflammation is the primary mechanistic link	Moderate	Consistent biomarker evidence across studies; MR shows IL-6 mediates COPD–CHD association; no single interventional trial confirms causality	(Goedemans et al., 2020; Terrosu, 2020; Yu et al., 2024; Roversi et al., 2014)
COPD is a direct independent cause of CHD	Moderate	MR study abolishes association after adjusting for inflammatory/lipid mediators; observational residual confounding by smoking; one angiographic study shows negative association	(Yu et al., 2024; Rothnie et al., 2015; Hong et al., 2019)
Lung hyperinflation is the strongest pulmonary predictor of CAD	Moderate	Two large smoker cohorts show hyperinflation attenuates FEV1/emphysema associations; cross-sectional design limits causal inference	(Chandra et al., 2021)

FIGURE 7 Key claims and supporting evidence identified in these papers

5. Conclusion

The literature establishes a strong, replicated, and clinically significant association between COPD and myocardial infarction/coronary artery disease that persists after adjustment for traditional cardiovascular risk factors in most observational analyses. The evidence for a causal contribution from COPD-specific mechanisms — particularly systemic inflammation, oxidative stress, and the acute inflammatory surge during exacerbations — is compelling, though the Mendelian randomization evidence suggests that the COPD–CHD causal pathway is substantially mediated by inflammatory and lipid pathways rather than representing a direct independent effect. The distinction between correlation and causation remains partially unresolved, hinging on the adequacy of smoking adjustment and the degree to which shared inflammatory biology represents confounding versus mediation.

Research Gaps

The evidence base contains notable gaps in causal inference, mechanistic specificity, and clinical translation that future research must address.

Topic/Outcome	RCT Evidence	Causal Inference (MR)	Mechanistic Specificity	Long-Term Outcomes	Clinical Generalization
COPD → MI risk	GAP	2	<research_gap		

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